NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

Assignment 2: Differential Amplifiers

Course Title: IC Design Lab (EC705)

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Branch: VLSI Design

Program Name: MTech (Research)

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Objective

This assignment aims to simulate and analyse the performance of differential amplifier circuits using matched nMOS and pMOS transistors. The objective is to understand their operating principles, evaluate key characteristics such as gain and explore various differential amplifier configurations as outlined in the subsequent questions.

Introduction

A **differential amplifier** is a fundamental building block in analog circuit design, primarily used to amplify the difference between two input signals while rejecting common-mode components. The objective of this simulation is to analyse the behaviour of a basic MOSFET differential amplifier and evaluate its key performance parameters, including differential gain response, and signal linearity.

Simulation Tool: Cadence Virtuoso.

Parameters

- Use gpdk180 nm file as model file.
- The width and Length of the MOSFET should be chosen to the accuracy of two decimal places only.
- L = 1 μm and W_{min} = 0.4 µm.
- The circuit should be operating on 1.8V.

QUESTIONS

1. **Design a Current Mirror loaded Differential Amplifier using 0.18 µm CMOS technology node using Cadence tools for the specifications mentioned in the table**.

Gain-Bandwidth	**≥ 5 MHz**
Av	**≥ 120**
C_L	**12 pF**
$ICMR_{max}$	**1.6 V**
$ICMR_{min}$	**1 V**
Slew rate	**6 V/ µsec**
Power dissipation	**< 0.5 mW**

Solution: -

Parameters:

$$t_{ox} = 4.0 \text{ nm}$$

$$\varepsilon_r = 3.9$$

$$\varepsilon_o = 8.85 \times 10^{-12} \text{ F/m}$$

$$\varepsilon_{ox} = \varepsilon_r . \varepsilon_o = 3.4515 \times 10^{-11} \text{ F/m}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = 8.62875 \times 10^{-3} \text{ F/m}^2$$

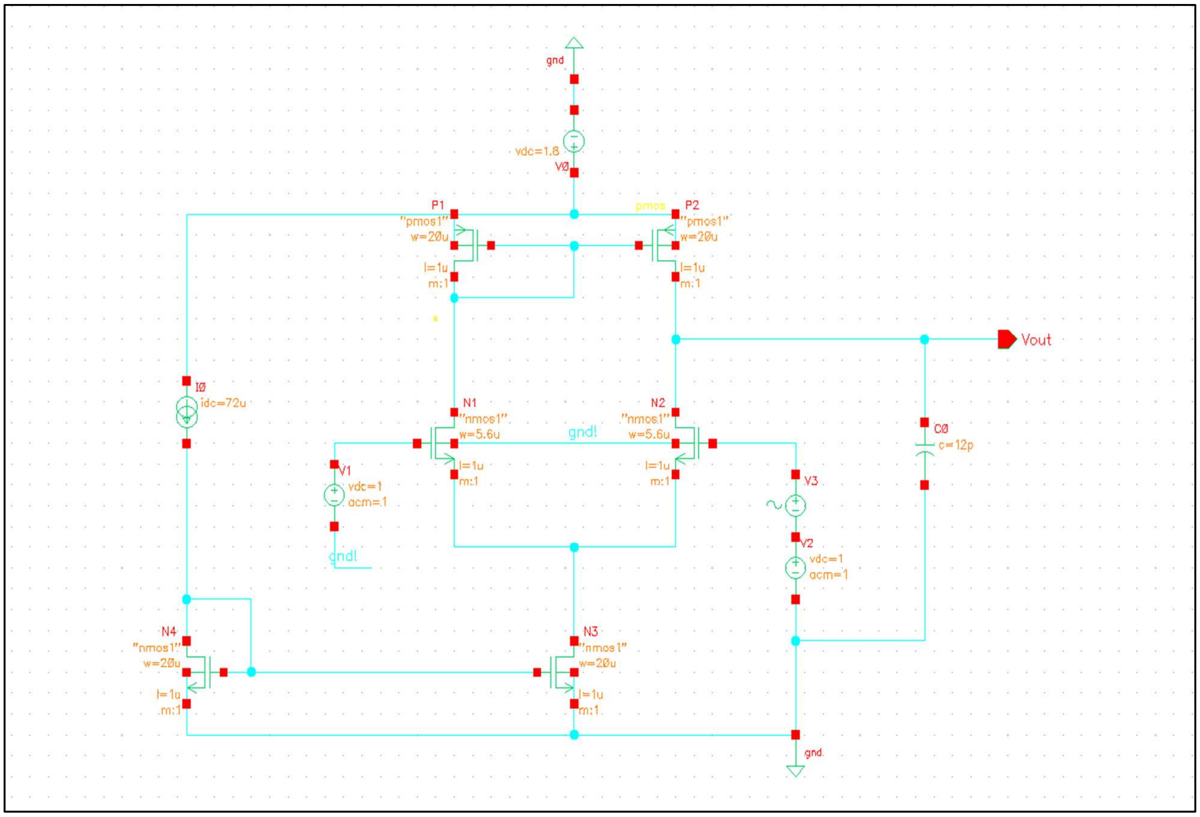
$$u_n = 0.046 \text{ m}^2\text{/Vs}$$

$$V_{thn} = 0.49 \text{ V}$$

$$u_p = 0.01 \text{ m}^2\text{/Vs}$$

$$V_{thp} = -0.45 \text{ V}$$

Schematic:



Meeting Specifications:

- Slew Rate = $\frac{dV_0}{dt} = 6 \times 10^6$ V/s

- Current Calculation:

$$I_C = C_L \frac{dV_0}{dt}$$

$$I_C = 6 \times 10^6 \times 12 \times 10^{-12} \text{ A}$$

$$I_C = 72 \text{ µA}$$

$$I_{D,\,N3} = 72 \text{ µA}$$

- **All the transistors must be in saturation.**

a) Considering N1 transistor,

$$V_{d,\,N1} \geq V_{g,\,N1} - V_{thn}$$

$$V_{d,\,N1} \geq V_{in} - V_{thn}$$

$$V_{d,\,N1} \geq ICMR_{max} - V_{thn}$$

$$V_{d,\,N1} \geq 1.6 - 0.49$$

$$V_{d,\,N1} \geq 1.11 \text{ V}$$

b) Considering P1 transistor,

$$V_{g,\,P1} = V_{d,\,P1} = V_{d,\,N1} = 1.11 \text{ V}$$

$$V_{s,\,P1} = 1.8 \text{ V}$$

$$V_{gs,\,P1} = V_{g,\,P1} - V_{s,\,P1}$$

$$V_{gs,\,P1} = 1.11 - 1.8$$

$$V_{gs,\,P1} = -0.69 \text{ V}$$

$$\text{So, } I_{DS,\,P1} = \frac{1}{2} u_p C_{ox} \frac{W_{P1}}{L_{P1}} (|V_{gs,\,P1}| - |V_{thp}|)^{2,}$$

We get $W_{P1} = W_{P2} \approx 20$ µm

c) Considering N1 transistor,

$$A_v = g_{m,\,N1}\,(r_{on} \parallel r_{op})$$

$$BW = \frac{1}{2\pi (r_{on} \parallel r_{op})\, C_L}$$

$$A_v \,.\, BW \geq 5 \text{ MHz}$$

$$\frac{g_{m,N1}}{2\pi C_L} \geq 5 \text{ MHz}$$

$$g_{m,\,N1} \geq 0.377 \text{ mA/V}$$

Consider $g_{m,\,N1} = 0.4$ mA/V

$$\text{So, } \frac{2 I_{d,N1}}{V_{gs,N1} - V_{thn}} = g_{m,\,N1}$$

$$V_{gs,\,N1} = V_{thn} + \frac{2 I_{d,N1}}{g_{m,N1}}$$

$$V_{gs,\,N1} = 0.67 \text{ V}$$

$$W_{N1} = W_{N2} = \frac{2I_{D,N1}\, L_{N1}}{\mu_n C_{ox}(V_{GS,N1} - V_{thn})^2}$$

Therefore, $W_{N1} = W_{N2} = 5.6\ \mu m$

d) Considering N2 transistor,

$$I_{DS,\ N2} = \frac{1}{2} u_n C_{ox} \frac{W}{L} (V_{gs,\ N2} - V_{thn})^2,$$

We get $V_{gs,\ N2} = 0.67$ V

$$V_{in} > V_{gs,\ N2} + V_{dssat,\ N3}$$

$$ICMR_{min} > V_{gs,\ N2} + V_{dssat,\ N3}$$

$$V_{dssat,\ N3} < ICMR_{min} - V_{gs,\ N2}$$

$$V_{dssat,\ N3} < 0.33\ V$$

Consider $V_{dssat,\ N3} = 0.3$ V

e) Consider N3 transistor,

$$I_{DS,\ N3} = \frac{1}{2} u_n C_{ox} \frac{W_{N3}}{L_{N3}} (V_{dssat,\ N3})^2$$

We get $W_{N3} = W_{N4} = 4\ \mu m$

Tabulation:

Width of transistors:

N_1, N_2	5.6 µm
N_3, N_4	4 µm
P_1, P_2	20 µm

N_3, N_4 transistor current mirror was giving current of 65 µA, so to get 72 µA width of N_3, N_4 was varied to 20 µm to get a current approximate to 71 µA, and N_1, N_2 was varied to 10 µm to get a gain of more than 120, whereas I got around 132.

- **DC Operating Points:**

1. N1

gbs	0
gds	1.31238u
gm	317.823u
gmb	83.8587u
gmbs	83.8587u
gmoveri	NaN
gmoverid	9.08447
go	NaN
i1	34.9853u
i3	-34.9853u
i4	-25.8811f
ib	NaN
ibd	-25.5655f
ibe	-1.40391p
ibs	-315.602a
ibulk	-25.8811f
id	34.9853u
idb	315.602a
ide	34.9853u

rout	761.973K
sdint	2.14748G
sdop	-1
self_gain	242.173
tau1	NaN
tk	NaN
trise	NaN
ueff	35.8645m
vbs	-261.727m
vdb	1.11631
vds	854.578m
vdsat	209.809m
vdsat_marg	NaN
vdss	209.809m
vearly	26.6579
vfbeff	-956.327m
vgb	1
vgd	-116.305m
vgs	738.273m

2. N2

gbs	0
gds	1.31238u
gm	317.823u
gmb	83.8587u
gmbs	83.8587u
gmoveri	NaN
gmoverid	9.08447
go	NaN
i1	34.9853u
i3	-34.9853u
i4	-25.8811f
ib	NaN
ibd	-25.5655f
ibe	-1.40391p
ibs	-315.602a
ibulk	-25.8811f
id	34.9853u
idb	315.602a
ide	34.9853u

rout	761.973K
sdint	2.14748G
sdop	-1
self_gain	242.173
tau1	NaN
tk	NaN
trise	NaN
ueff	35.8645m
vbs	-261.727m
vdb	1.11631
vds	854.578m
vdsat	209.809m
vdsat_marg	NaN
vdss	209.809m
vearly	26.6579
vfbeff	-956.327m
vgb	1
vgd	-116.305m
vgs	738.273m

3. N3

gbs	0
gds	13.6653u
gm	867.235u
gmb	248.663u
gmbs	248.663u
gmoveri	NaN
gmoverid	12.3943
go	NaN
i1	69.9707u
i3	-69.9707u
i4	-1.05877f
ib	NaN
ibd	-1.05877f
ibe	-262.786f
ibs	0
ibulk	-1.05877f
id	69.9707u
idb	1.05838f
ide	69.9707u

rout	73.1782K
sdint	2.14748G
sdop	-1
self_gain	63.4628
tau1	NaN
tk	NaN
trise	NaN
ueff	46.8439m
vbs	0
vdb	261.727m
vds	261.727m
vdsat	148.9m
vdsat_marg	NaN
vdss	148.9m
vearly	5.12033
vfbeff	-914.867m
vgb	584.153m
vgd	322.426m
vgs	584.153m

4. N4

gds	3.91338u
gm	896.788u
gmb	257.052u
gmbs	257.052u
gmoveri	NaN
gmoverid	12.4554
go	NaN
i1	72u
i3	-72u
i4	-1.06523f
ib	NaN
ibd	-1.06523f
ibe	-585.218f
ibs	0
ibulk	-1.06523f
id	72u
idb	1.05838f
ide	72u

rout	255.534K
sdint	2.14748G
sdop	-1
self_gain	229.16
tau1	NaN
tk	NaN
trise	NaN
ueff	46.8446m
vbs	0
vdb	584.153m
vds	584.153m
vdsat	148.955m
vdsat_marg	NaN
vdss	148.955m
vearly	18.3984
vfbeff	-914.942m
vgb	584.153m
vgd	0
vgs	584.153m

5. P1

gbs	0
gds	1.57875u
gm	267.74u
gmb	87.5853u
gmbs	87.5853u
gmoveri	NaN
gmoverid	7.65187
go	NaN
i1	-34.9902u
i3	34.9902u
i4	51.7469f
ib	NaN
ibd	51.7469f
ibe	735.442f
ibs	-0
ibulk	51.7469f
id	-34.9902u
idb	51.7361f
ide	-34.9902u

rout	633.414K
sdint	2.14748G
sdop	-1
self_gain	169.591
tau1	NaN
tk	NaN
trise	NaN
ueff	7.73804m
vbs	0
vdb	-683.695m
vds	-683.695m
vdsat	-214.609m
vdsat_marg	NaN
vdss	-214.609m
vearly	22.1633
vfbeff	-950.682m
vgb	-683.695m
vgd	0
vgs	-683.695m

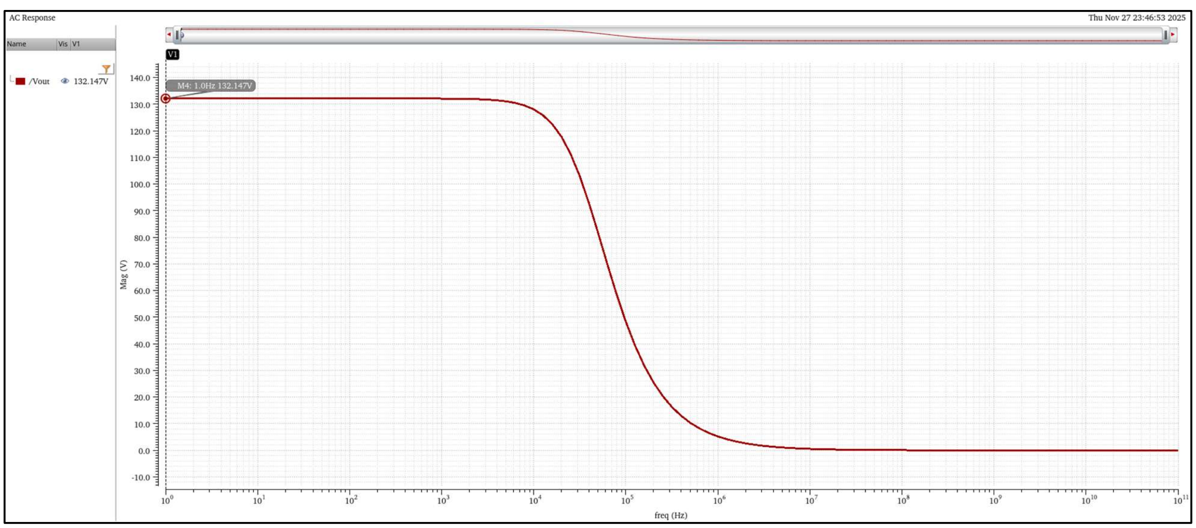
6. P2

gbs	0
gds	1.57875u
gm	267.74u
gmb	87.5853u
gmbs	87.5853u
gmoveri	NaN
gmoverid	7.65187
go	NaN
i1	-34.9902u
i3	34.9902u
i4	51.7469f
ib	NaN
ibd	51.7469f
ibe	735.442f
ibs	-0
ibulk	51.7469f
id	-34.9902u
idb	51.7361f
ide	-34.9902u

rout	633.414K
sdint	2.14748G
sdop	-1
self_gain	169.591
tau1	NaN
tk	NaN
trise	NaN
ueff	7.73804m
vbs	0
vdb	-683.695m
vds	-683.695m
vdsat	-214.609m
vdsat_marg	NaN
vdss	-214.609m
vearly	22.1633
vfbeff	-950.682m
vgb	-683.695m
vgd	444.089a
vgs	-683.695m

Operating Points	N_1	N_2	N_3	N_4	P_1	P_2
$g_{m,\,eff} = g_m + g_{mb}$	401.68 µS	401.68 µS	1115.90 µS	1153.84 µS	355.33 µS	355.33 µS
r_{out}	761.973 Ω	761.973 Ω	73.1782 Ω	255.534 Ω	633.414 Ω	633.414 Ω
I_d	34.9853 µA	34.9853 µA	69.9707 µA	72 µA	34.9902 µA	34.9902 µA
V_{gs}	0.738273 V	0.738273 V	0.738273 V	0.584153 V	−0.683695 V	−0.683695 V
V_{ds}	0.854578 V	0.854578 V	0.261727 V	0.584153 V	−0.683695 V	−0.683695 V
V_{dsat}	0.209809 V	0.209809 V	0.148900 V	0.148955 V	−0.214609 V	−0.214609 V

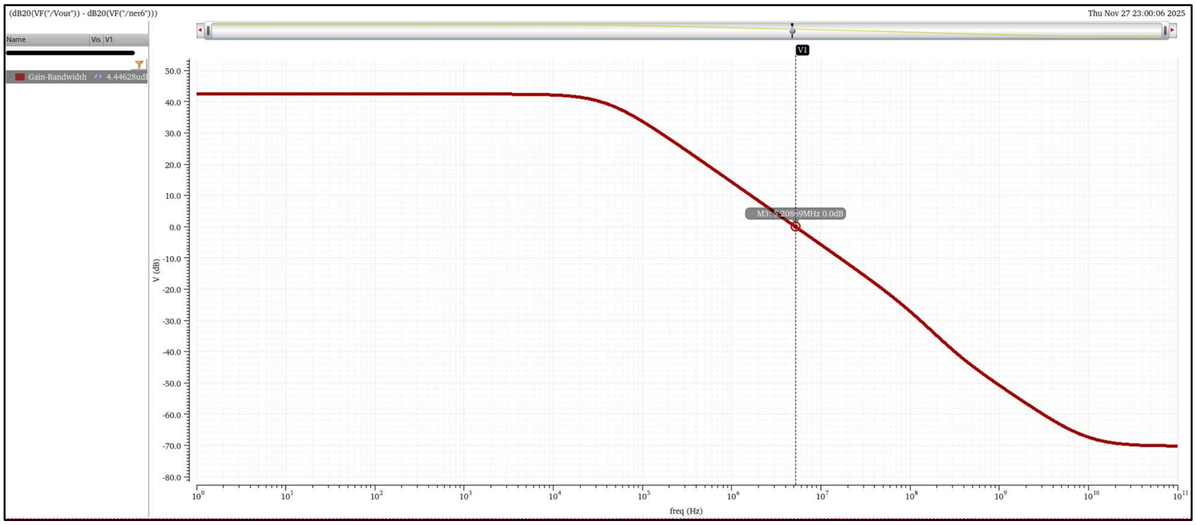
- **Gain (at $ICMR_{min}$)**



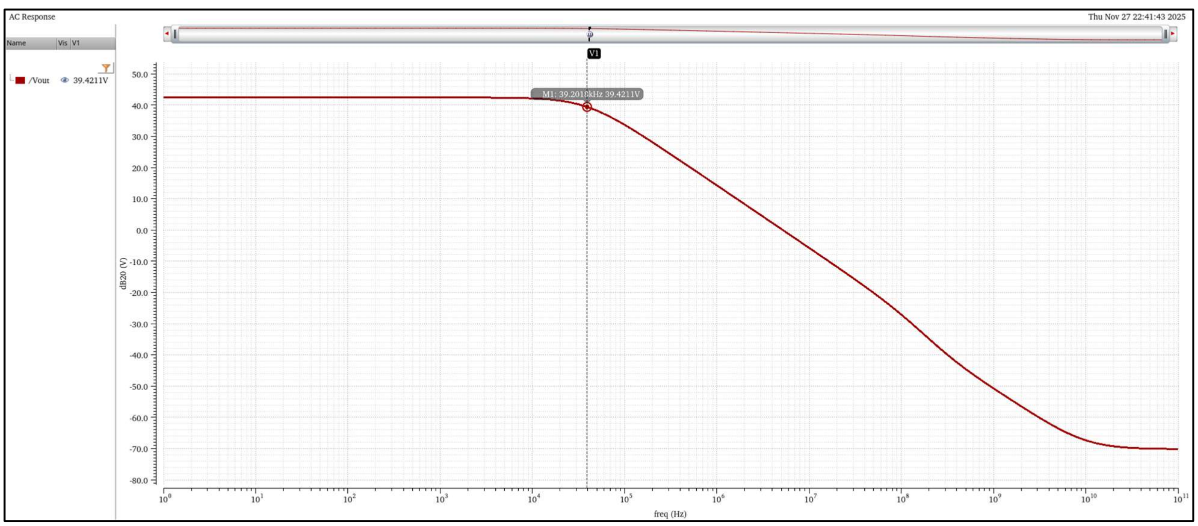
- **Slew Rate (at $ICMR_{min}$)**

	Expression	Value
1	slewRate(VT("/V...	3.750E6

- **Gain-Bandwidth (at ICMR$_{min}$)**

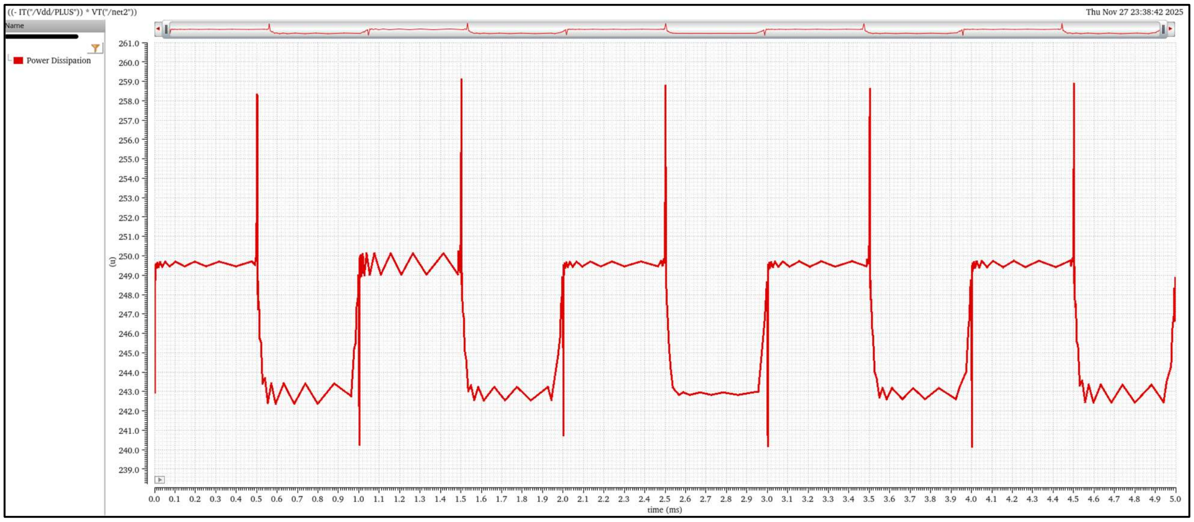


- **Bandwidth (at ICMR$_{min}$)**

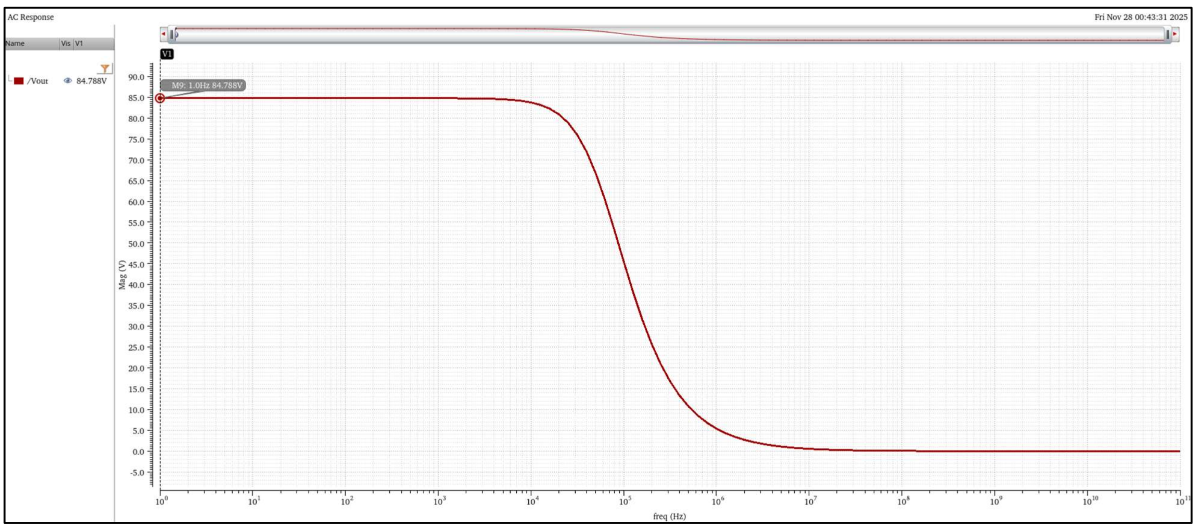


- **Power Dissipation (at ICMR$_{min}$)**

	Expression	Value
1	Power dissipation...	246.5u



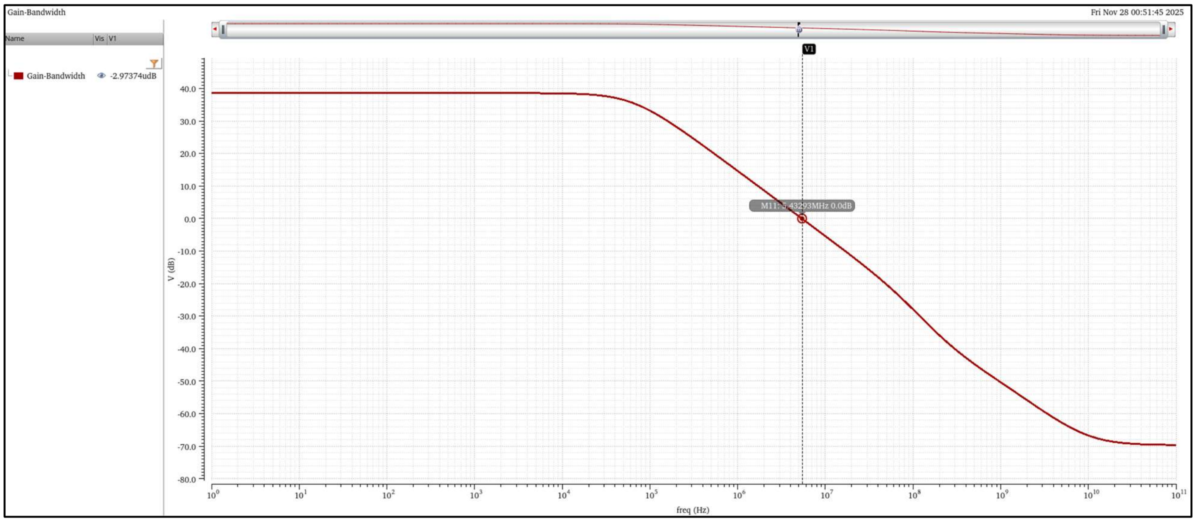
- **Gain (at $ICMR_{max}$)**



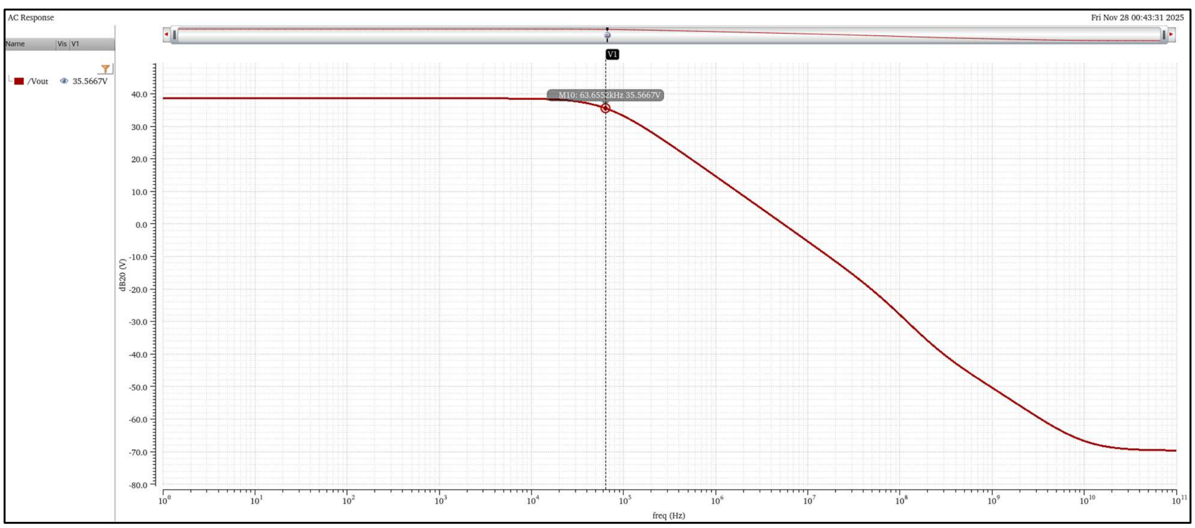
- **Slew Rate (at $ICMR_{max}$)**

	Expression	Value
1	slewRate(VT("/Vout...	2.100M

- **Gain-Bandwidth (at ICMR$_{max}$)**

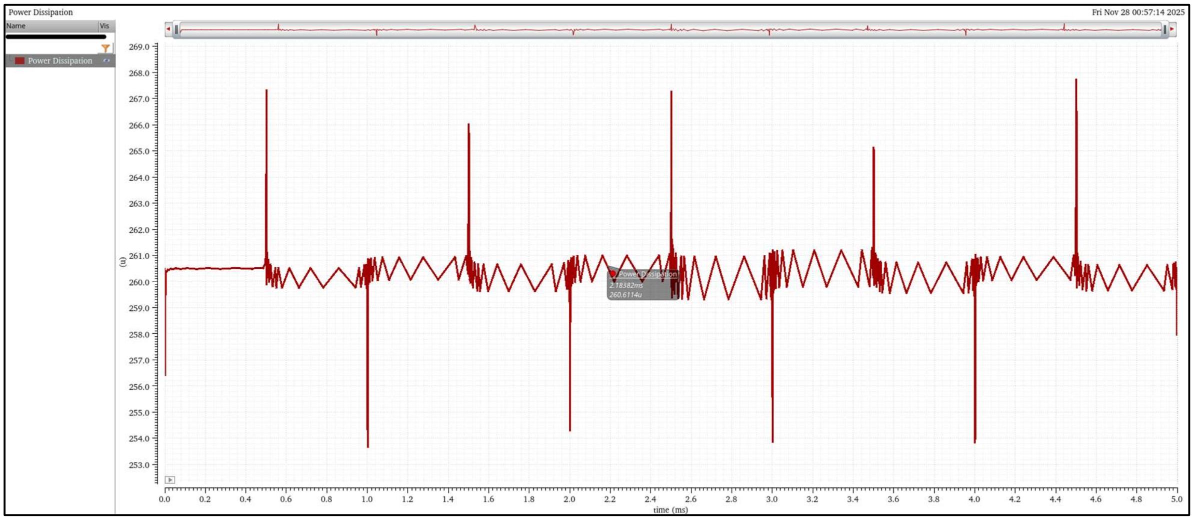


- **Bandwidth (at ICMR$_{max}$)**



- **Power Dissipation (at ICMR$_{max}$)**

	Expression	Value
1	Power Dissipation ...	246.5u



Obtained Results:

Parameters	$ICMR_{min}$	$ICMR_{max}$
Gain	132.147	84.788
Slew Rate	3.75 V/μA	2.1 V/μA
Gain-Bandwidth	5.208 MHz	5.433 MHz
Bandwidth	39.201 kHZ	63.655 kHz
Power Dissipation	246.5 uW	246.5 uW

Conclusion: Current Mirror Loaded Differential Amplifiers are designed with given specifications

---End of Report---